



AGENT CTOS: ON-CHAIN
SOLUTIONS

Agent CTOS: On-chain Underwriting for Agents

On-chain underwriting for agents that pay strangers · Live on Solana devnet

69,000

AI agents already settling
payments on x402

**\$300-500
billion**

Estimated US market share
by 2030

165 million

Agent payments settled to
date

Solution



Score

A merchant's score is derived solely from payments they have actually completed — nothing else. No identity. No KYC. No off-chain data.

Instant vs. Escrow

A trusted merchant is paid instantly. An unknown one has funds held in escrow until delivery is confirmed. Nobody is ever blocked or banned — the worst outcome is a wait.

Self-Recovery

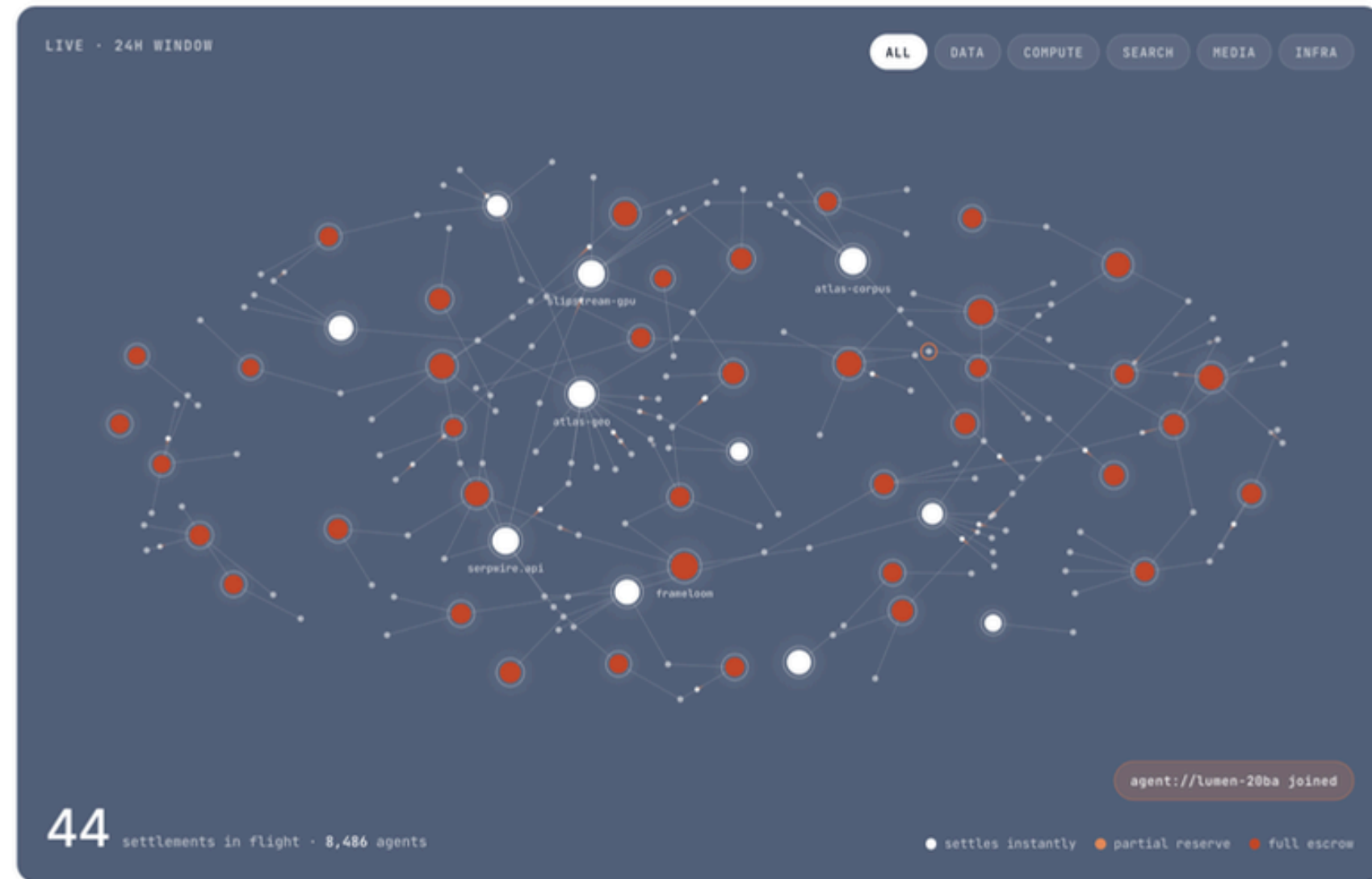
If a merchant never delivers, the buyer reclaims the escrowed funds themselves after timeout. No support ticket. No dispute team. No company in the middle.

Demo

GLOBAL NETWORK

Every settled payment, drawn as it happens.

Agents on the outside, merchants at the hubs, one line per settlement relationship. Node colour is the escrow a merchant's next order would require — nothing else.



<https://agent-ctos.vercel.app>

Market, Why Now, Why Solana



Market Size

Agent-driven commerce grows from \$8B in 2026 to \$1.5T globally by 2030. US market alone: \$300–500B.

Source: Juniper Research, April 2026; Bain & Company, December 2025

Why Now

The x402 Foundation launched under the Linux Foundation, backed by Visa, Mastercard, and Ripple. Payment rails for agents are being standardised right now.

Source: CoinDesk, July 2026

Why Solana

EVM has the Commerce Payments Protocol. Solana's x402 scheme has no native escrow or commerce spec. We are that missing piece — and it must live on-chain so the score is verifiable by anyone.

Alternatives

Commerce Payments Protocol and x402r on EVM: same concept, wrong chain for this gap

Facilitators with private risk tables: faster to build, but scores cannot be verified or carried elsewhere

Traditional processors with rolling reserves: proven, but they ban merchants and keep data to themselves

Why we win

Portability: reputation earned at one facilitator is honoured by every other

No bans: a false positive costs a delay, not a business

Verifiable on-chain score: any participant can audit the underwriting, not just the facilitator

Solana-native: no equivalent escrow or commerce spec exists on x402 today

Go to Market

01

Month One: Claim and Reserve

Ship merchant fulfilment claim, onboard relationship accounts, and deploy the rolling reserve mechanism. Establish baseline scoring from completed payments.

02

Month Two: Facilitator Shim

Run behind one x402 facilitator's verify and settle calls as a drop-in shim. Zero integration lift for the facilitator. Score every payment silently.

03

Month Three: MCP Distribution

Distribute through MCP to agent developers. Publish leverage ratio and loss rate publicly. One integration reaches every agent built on the framework.

04

Distribution Strategy

Distribution is through agent frameworks, not merchant sales. We do not sell to merchants. We reach them through the developers who build the agents that pay them.

Reserve Region

Credit covered by the merchant's own posted reserve
— extraction is impossible

Leverage Gap

Credit extended above collateral — our chosen loss
rate, bounded by reserve, not identity

Sybil Proof

Fake reputation unlocks nothing beyond the reserve —
worthless by construction

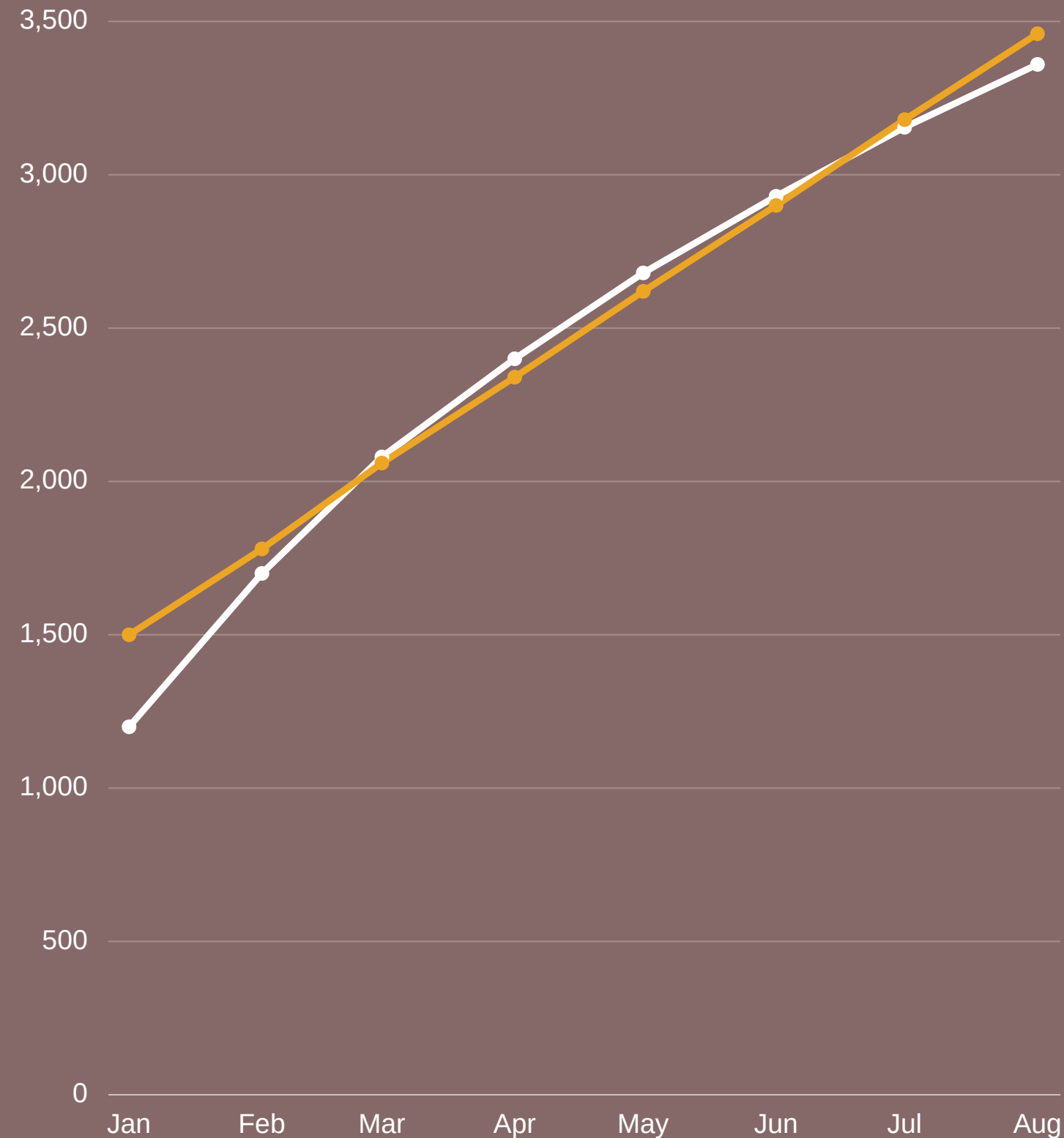
Appendix: How a Payment Routes

Merchant Reserve → None
No buyer history: Full escrow
Established buyer: Full escrow
Merchant Reserve → Partial
No buyer history: Full escrow
Established buyer: Full escrow
Merchant Reserve → Covers exposure
No buyer history: Full escrow
Established buyer: Instant settle ✓

Decision logic

Only one cell settles instantly: an established buyer paired with a merchant reserve that fully covers the exposure. Every other combination routes to full escrow — funds are held until delivery is confirmed or the timeout expires. Throwaway wallets have nothing to gain: without buyer history, even a fully reserved merchant triggers escrow. Fake wallets cannot extract value by construction. The score alone is never sufficient — collateral must be present. Vertical axis: merchant reserve (none / partial / covers exposure)
Horizontal axis: buyer standing (no history / established)

Faking Reputation Stops Paying



Source: AI-generated data. Replace or verify before use.

Payoff vs. Cost of Faking Reputation

Source: Agent CTOS model. Payoff = $k\sqrt{V}$; Cost = cV . Crossover defines maximum extractable value.

Payoff grows as the square root of counterparty-diverse volume. Cost grows linearly with faked volume. The largest gap between the two curves is the maximum extractable value — a constant we choose deliberately. Past the crossover, every additional faked transaction loses money.

Appendix: Attacks & Answers

01

The Rug

Escrow holds the funds. The buyer reclaims after timeout with no merchant signature required. No support ticket, no dispute team, no intermediary.

02

Manufactured Reputation

A perfect fake history unlocks credit only up to the posted reserve. Extraction is bounded by collateral, not identity. Fake reputation is worthless by construction.

03

Run for the Exit

Size the reserve against peak concurrent exposure, never trailing volume. A merchant cannot outrun their collateral obligation at the moment of maximum risk.

04

Dispute Swarms

A buyer with no history receives full escrow on every payment. There is nothing to dispute and nothing to extract. Throwaway wallets have no attack surface.

What we do not fix

- A patient attacker who earns real reputation then rugs extracts the leverage gap
- That gap is our loss rate: chosen deliberately, and published
- An honest merchant disputed by an honest but mistaken buyer still loses

Our position

- We price fraud, we do not prevent it
- This is hackathon-grade and unaudited
- Residual risks are stated before anyone asks, not discovered after
- Transparency about limits is part of the design, not a disclaimer

References for Research and Insights

Juniper Research, Agentic Commerce Set to Generate \$1.5 Trillion Globally by 2030, 7 April 2026. juniperresearch.com/press

Bain & Company, 2030 Forecast: How Agentic AI Will Reshape US Retail, December 2025. bain.com/insights

Morgan Stanley, Agentic Commerce Impact Could Reach \$385 Billion by 2030. morganstanley.com/insights

Additional Relevant Studies and Reports

MarketsandMarkets, AI Agents Market Global Forecast to 2030. marketsandmarkets.com

Chainalysis, Inside x402: 100M Agentic Payments on Base. chainalysis.com/blog

CoinDesk, Visa, Mastercard and Ripple join the standard letting AI agents pay in stablecoins, 15 July 2026. coindesk.com